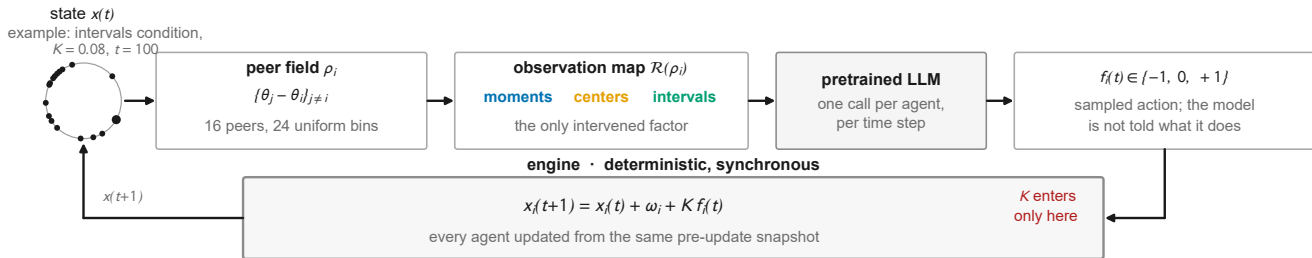


a Engine and information-flow contract



Withheld from the model at every call

common action/output contract; map-specific observation description and payload

coupling K

absolute phase

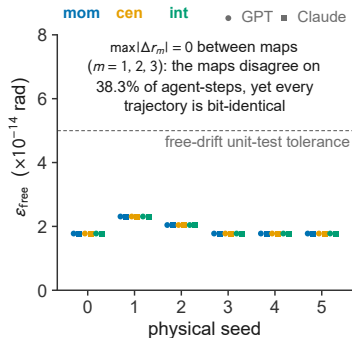
agent identity

time step t

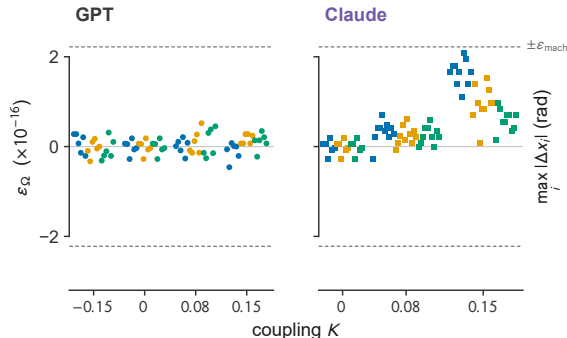
previous actions

trajectory history

b Exact $K = 0$ free evolution



c Collective-frequency accounting identity



d Synchronous-update audit

